

NISHAL STANISLAUS SILVA, PhD

Computer Vision & ML Engineer | Industrial Inspection, IoT & Edge Deployment

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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PROFESSIONAL SUMMARY

Computer Vision and ML Engineer with 5.5 years building and deploying production machine-vision and automation systems for manufacturing at MAS Holdings, South Asia's largest apparel manufacturer, plus a PhD in real-time embedded ML. Shipped an automated multilingual care-label QC system (OpenCV template and feature matching, conveyor with PLC-controlled reject) that cut inspection time 99.5%, and a loom-side fabric structural-defect detection system (microscopic camera array) that cut operator headcount 50%. Comfortable owning computer-vision R&D end to end: algorithm design for pattern recognition and OCR, camera and sensor integration, and reverse-engineering PLC and electro-mechanical trigger points to close the loop between vision and automation hardware. C++/Python systems engineer with a track record of taking a system from prototype to a production floor with a human-in-the-loop path for edge cases, and of training operators and support engineers on the result.

CORE COMPETENCIES

Computer Vision & Perception: OpenCV (template and feature matching, image registration and warping, corner/edge detection), OCR and layout segmentation, industrial and microscopic camera-array systems, explainable defect overlays for human-in-the-loop QC, multilingual label and print inspection, projector-camera AR alignment.

ML Engineering: TensorFlow, Keras, PyTorch, scikit-learn, pattern-recognition algorithm design, benchmark and ablation-driven evaluation.

Systems & Deployment: C++/Python real-time inference engines, PLC integration and electrical/software co-design for automation systems, high-frequency IoT sensor ingestion and ETL, customer-facing deployment, commissioning support, and technical training delivery.

TECHNICAL STACK

Languages: C++, Python, C, MATLAB, JavaScript/TypeScript, Shell.

ML / AI: TensorFlow, Keras, PyTorch, scikit-learn.

Computer Vision: OpenCV, image registration and warping, corner/edge detection, industrial camera arrays, microscopic imaging.

Data & Cloud: AWS, SQL, Pandas/NumPy, ETL pipelines.

Dev & MLOps: Git, Docker, CI/CD, pytest.

EDUCATION

PhD, Information Engineering and Computer Science – University of Trento, Italy 2021 - 2025

MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK 2016 - 2018

BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK 2013 - 2014

PROFESSIONAL EXPERIENCE

Research Engineer | MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka Jan 2015 - Jul 2020

- Designed and deployed end-to-end computer-vision and IoT systems for manufacturing at scale, taking camera and sensor streams directly into production and automation workflows; drove operational efficiency gains of up to 300%.

- Built an automated multilingual care-label QC system using OpenCV template and feature matching with an explainable defect overlay, iterating it from a handheld scanner to an industrial camera on a conveyor with PLC-controlled reject, cutting inspection time 99.5% with a human-in-the-loop path for borderline cases.
- Developed a loom-side fabric structural-defect detection system using a microscopic camera array with overlapping fields of view, cutting operator headcount 50%.
- Reverse-engineered non-invasive PLC trigger points and RIP automation hooks for the Aeoon Kyo direct-to-garment printer, coordinating three closed-loop subsystems in custom C++; separately designed the geometry-alignment algorithm and built the RIP integration for Promptly, MAS/Twinery's on-demand two-sided garment printing product, now deployed in the US, Mexico, France and Sri Lanka.
- Ran hands-on computer-vision training seminars and exhibited solutions at inter-company expos supporting multi-factory rollouts; introduced CI/CD and code-review practice and mentored engineers.

Postdoctoral Researcher | *University of Trento, Trento, Italy*

Jan 2025 - Dec 2025

- Owned the full ML pipeline for streaming audio and IMU/gesture data: acquisition, DSP/feature engineering, training and evaluation, and edge deployment on Raspberry Pi 4, with monitoring.
- Designed frozen-protocol benchmark suites and ablation studies quantifying accuracy/latency/CPU trade-offs, the same discipline used to validate the QC systems above against production tolerances.

Visiting Researcher | *McGill University, IDMIT/CIRMMT, Montreal, Canada*

May 2024 - Aug 2024

- Built a self-powered smart electric guitar (BNO055 IMU + HiFiBerry + Raspberry Pi 4), fusing IMU and audio signals via a hierarchical finite-state machine, achieving F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka*

Aug 2020 - Feb 2021

- Built ML and statistical pipelines for financial forensics, compliance monitoring, and anomaly detection on large structured enterprise data.

SELECTED PROJECTS (full portfolio: nishal.xyz/research)

- **Tech-Pack Digitization:** Pre-LLM OCR and layout segmentation of physical garment blueprints into a searchable digital archive.
- **Lace Archive & ERP Search:** Flatbed and microscopic capture preserving lace relief detail with designer metadata tagging and context-aware ERP search; still in production use.
- **Nebula:** open-source C++17 audio-feature-extraction library with per-feature ARM latency benchmarks (*nebula on GitHub*)

PUBLICATIONS

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *Journal of the Audio Engineering Society*, 74(3). Silva, N. & Turchet, L. (2022). A Structural Similarity Index Based Method to Detect Symbolic Monophonic Patterns in Real-Time, achieving 95% detection without training data. *Proceedings of the International Conference on Digital Audio Effects (DAFx)*.

HIGHLIGHTS

MIDI Innovation Awards 2023 Finalist ("Hot Licks", Prototypes & Non-Commercial Software). GMOA recognition (Government Medical Officers' Association, Sri Lanka, 2020) for a COVID-19 remote patient-vitals computer-vision monitoring deployment. Uses AI-assisted engineering workflows including Claude Code for tooling and pipeline development.